Knowledge-Based Systems in Bioinformatics

Lab 5 – Monte Carlo feature selection

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Introduction

Monte Carlo feature selection (MCFS) is an algorithm for feature selection and attribute ranking. The algorithm is well described in the article **Monte Carlo feature selection for supervised classification** (Bioinformatics, 2008) by Michal Draminski, Alvaro Rada Iglesias, Stefan Enroth, Claes Wadelius, Jacek Koronacki and Jan Komorowski.

In this exercise, we are using the same dataset as in the article to reproduce the results and get acquainted with MCFS. The data set contains microarray lymphoma data from Alizadeh *et al* (Nature, 2000). There are gene expression (4026 genes) samples from 62 patients: 42 with lymphoma and leukemia (D), 9 with follicular lymphoma (F) and 11 with chronic lymphocytic leukemia (C). The task is to identify genes that show differences in expression between the three different classes. As normal using microarray experiments we have a dataset with thousands of attributes (genes) and few objects (patients). For this type of problems, the feature selection can actually be the hardest and most important task.

rmcfs is an implementation of the MCFS algorithm in R. You can find documentation at:
https://cran.r-project.org/web/packages/rmcfs/rmcfs.pdf

An introduction to R:
https://cran.r-project.org/doc/manuals/r-release/R-intro.pdf

Please note that R 2.7 or higher and Java 6.0 or higher are required by *rmcfs*.

What is more, the agreement between the R and Java version is required i.e. 64-bit or 32-bit.

For macOS use command `sudo R CMD javareconf` in Terminal

To change the R version in RStudio on computers with a Windows operating system please follow the link:

https://support.rstudio.com/hc/en-us/articles/200486138-Changing-R-versions-for-RStudio-desktop

Reopen R or Rstudio and install *rjava* and *rmcfs* (install it only once):

```
install.packages("rJava")
install.packages("rmcfs")
```

then load the package. Remember that you have to read the library each time you run R or Rstudio:

```
library(rJava)
library(rmcfs)
```

The macOS tutorial how to set up rJava. Try with Java version 11:

https://zhiyzuo.github.io/installation-rJava/

Tasks

1. Load the Alizadeh dataset.

```
alizadeh <- read.table("file_path/alizadeh.csv")
```

where *file_path* is the directory where you have downloaded the dataset to.

Display the dataset using

`alizadeh[1:n, c(1:n, dim(alizadeh)[2])],`

where `n` is the number of lines do display

Task: Reveal the structure of the given dataset: what are the set of attributes, decision attribute and the universe?

Task: Check alizadeh$class and tell what is the number of objects in each class. You can use a `table()` function.

2. Run *mcfs* function by:

```
result <- mcfs(class~., alizadeh, projections=1500,
projectionSize=0.1, splits=5, splitSetSize=500,
cutoffPermutations = 6, threadsNumber = 8)
```

Task: Explain the function of parameters *projections, projectionSize, split, cutoffPermutations* and *splitSetSize*.

Task*:* By typing `head(result$RI)` you should see 6 columns of the results table. Describe the results table.

3. Plot the "distance graph".

```
plot(result, type="distances")
```

You will see the "distance graph" showing the difference between the projections on the X-axis and ranks on the Y-axis. 0 corresponds to no difference. After a sufficient number of projections (depending on the settings and data set), the rankings will stabilize.

Task: How many projections are needed to stabilize our rankings? What does the graph indicate? Is it necessary to re-run the algorithm? If yes, what parameters should be changed?

4. The following figure shows an example of the confusion matrix for one permutation.

```
Confusion Matrix
                System Answer
        D       F       C       other
D       100274  21578   28148   0
F       25913   5944    8143    0
C       26209   6131    7660    0
other   0       0       0       0


Accuracy = 0.4951
WeightedAccuracy = 0.3361
True Positive Rate
        D: 0.6684
        F: 0.1486
        C: 0.1915
False Positive Rate
        D: 0.6515
        F: 0.1458
        C: 0.1910
```

Task: Make sure that you understand the confusion matrix. Next, find out how the accuracy, true positive rate and false positive rate are calculated.

5. The results are located under the `result` variable. The access to the results of mcfs can be obtained by typing `result$`.

```
result2 <- result$RI[1:result$cutoff_value,]
```

Task: List the significant features, i.e. gene names from the table, and their rankings. Compare your ranking to the ranking in the original article (p. 114).

6. To find out the importance of significant attributes, look into the variable `result2` which contains information for each attribute.

Recall that attributes with their RI_{norm} value above the significance cutoff are considered to be significant (p = 0.05), whereas the ones below the cutoff are insignificant. The values will always be in the interval 0 to 1, where higher values correspond to better classification ability.

Task: Which was the highest ranked insignificant gene? How big is the difference between the RI_{norm} for this gene and the lowest ranked significant gene?

7. Build a graph to show interdependencies between the top features using:

```
gid <- build.idgraph(result, size = 20)
plot.idgraph(gid, label_dist = 0.3)
```

Task: What is the interpretation of this graph?